Econometrics Part 2

The linear model in practice

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Outline

1. Multicollinearity
2. Handling categories
3. Omitted variables
4. Predictions in logs
5. Influential observations

Multicollinearity

- This problem arises if some variable is equal to an exact linear combination of some other variables (e.g. if we have variables like *income*, *income after tax* and *tax*)
- This variable is unnecessary because it contains redundant information
- Worse, it prevents the computation of the OLS estimator, since X is no longer full rank and we cannot compute $\hat{b} = (X'X)^{-1}X'y$
- Some software (e.g. Stata) detects it and arbitrarily removes one redundant variable
- statsmodels does **not**: it uses a pseudo-inverse and only prints a warning in the summary ("the design matrix is singular"), so always read it

Near multicollinearity

- This problem arises if some variable is *almost* equal to an exact linear combination of some other variables
- In other words, it means that they are very strongly correlated
- We *can* compute $\hat{b} = (X'X)^{-1}X'y$
- But: $\hat{b}$ is very unstable and the estimator is not very reliable (large variance)

How to detect near multicollinearity (1)

- The VIF (Variance Inflation Factor) indicates by how much the variance of a parameter $\hat{b}_j$ is inflated due to the correlation of variable X_j with all the other explanatory variables
- Say our model has k parameters: constant, X_1, X_2, ..., X_{k-1}
- We run the regression of each X_j on all the other explanatory variables, and save its R^2, called R_j^2
- Example: $X_1 = a_1 + a_2 X_2 + \cdots + a_{k-1} X_{k-1} + \varepsilon$
- Next, the VIF of each variable j is computed as:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

How to detect near multicollinearity (2)

- One can prove that in the original regression $y = b_0 + b_1 X_1 + \cdots + b_{k-1} X_{k-1} + u$:

$$V(\hat{b}_j) = \frac{\sigma^2}{(N-1)\,\widehat{Var}(X_j)}\,\text{VIF}_j$$

- $\text{VIF}_j > 5$ (some use 10) indicates a high risk of multicollinearity between variable j and the other variables
- Python: `from statsmodels.stats.outliers_influence import variance_inflation_factor`, then `variance_inflation_factor(X.values, j)`
- Careful: X must contain the constant column (`sm.add_constant`), otherwise the VIFs are wrong
- If you don't see any big issue, do not remove variables: risk of omitted variable bias (see below)

Dummy variables: 2 categories

- Let's say we have N individual observations providing income, gender and the number of years of education
- We want to explain individual income y by education x and gender z: $y_i = a + bx_i + cz_i + u_i$
- z is a *categorical variable*, ideally coded with 0/1
- Say $z = 1$ codes for males
- c is the average income difference between males and females with the same level of education
- It would be irrelevant to have both a male *and* a female dummy: since male + female = 1 = the constant, it would cause *exact multicollinearity* (the "dummy variable trap")
- Remark: having a categorical variable as the dependent variable requires other models, such as Probit and Logit (see later chapters)

Dummy variables: more than 2 categories

- We want to explain life satisfaction y by income x and type of environment z: *big city*, *small city* or *rural* (coded 1, 2 and 3)
- z is still a *categorical variable*, but with 3 categories: we choose a reference category, with respect to which parameters are computed, say *rural*
- Model: $y_i = a + bx_i + c_1 z_{1i} + c_2 z_{2i} + u_i$, where z_1 and z_2 are dummies for *big city* and *small city*
- Do not keep the original coding (1, 2, 3), which would be meaningless in a regression
- pandas: `pd.get_dummies(df["environment"], dtype=int)`, then drop the *rural* column (`drop_first=True` would drop *big city*, the first category)
- statsmodels formula: `C(environment, Treatment(reference=3))`

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Dummy variables: outliers

- Let's say we have N yearly observations for a country that experienced a shock
- We want to explain life expectancy y by the number of doctors per capita x: $y_t = a + bx_t + u_t$
- Let's say that the shock took place in year t_0: this unusual observation is likely to bias the whole regression, but we don't want to delete it
- We will take this event into account with a dummy variable:
- $D_t = 1$ if $t = t_0$, $D_t = 0$ otherwise
- The model is now written: $y_t = a + bx_t + cD_t + u_t$
- c represents the impact of the shock
- Note: the residual of year t_0 becomes 0, so a and b are estimated as if year t_0 were dropped

Omitted variable bias

- Estimating the previous model with and without dummy D can lead to changes even in parameters a and b
- Omitting a relevant variable (here D) that is correlated with the included variables leads to *omitted variable bias*, which affects all estimated parameters
- Proof: in the finite sample case with the Frisch-Waugh theorem (see Dormont), and proof involving consistency in later chapters (on endogeneity)
- This is a very important issue in applied work, so try not to forget important variables in models

The Frisch-Waugh theorem

- Let Z be a matrix of variables
- $M_Z = I - Z(Z'Z)^{-1}Z'$ is the orthogonal projection matrix on $L^{\perp}(Z)$
- Let's define 2 models:
  1. $y = Xb_{(i)} + Zc + u$
  2. $M_Z y = M_Z X b_{(ii)} + M_Z u$
- The Frisch-Waugh theorem states that $\hat{b}_{(i)} = \hat{b}_{(ii)}$
- Important consequence: estimating the model with both sets of variables X and Z **or** with X alone will in general not provide the same results
- This is called the "omitted variable bias"

Under which condition will the results be the same?

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Answer

- Estimating the model with both sets of variables X and Z **or** with X alone will yield the same results only if X and Z are orthogonal
- i.e. $X'Z = 0$ (for centered variables: X and Z are totally uncorrelated)
- This rarely happens with observational data (it is the idea behind randomized experiments)

A side effect of the FW theorem: centered variables

- Estimating the simple model $y_i = a + bx_i + u_i$ with or without centering observations (i.e. replacing each value y_i by $y_i - \bar{y}$, same for x) will give the same result for $\hat{b}$
- Why: apply the FW theorem with $Z =$ the constant, since M_Z simply centers the variables
- Centering observations will make the constant unnecessary

Advantages of using logs

1. Taking the *log* of the dependent variable can make the residuals look more "normal"
2. It can also reduce heteroskedasticity
3. Parameters can be interpreted as percentages (semi-elasticities) or, if x is also in logs, as elasticities

About the log transformation

- The average predicted y should be equal to the average observed y
- Here, we can compute a prediction of $\log(y)$
- Can we get back to the predicted y (in levels) simply by taking the exponential of the predicted $\log(y)$?

Log and predictions (1)

- Let's use model $\forall i,\ \log(y_i) = a + bx_i + u_i$ (what matters is that we take the *log* of y)
- Taking the exponential:
  $y_i = \exp(a + bx_i + u_i) = \exp(a + bx_i)\exp(u_i)$
- So if the u's are independent from the x's, we get:
  $E[y_i] = E[\exp(a + bx_i)\exp(u_i)] = E[\exp(a + bx_i)]\,E[\exp(u_i)]$
- We usually consider the x's as fixed, so:
  $E[y_i] = \exp(a + bx_i)\,E[\exp(u_i)]$
- By Jensen's inequality, $E[\exp(u_i)] > \exp(E[u_i]) = 1$: simply taking the exponential underestimates y
- A prediction for y is thus: $\hat{y}_i = \exp(\hat{a} + \hat{b}x_i)\,\hat{E}[\exp(u_i)]$
- But how to estimate $E[\exp(u_i)]$ (the retransformation parameter)?

Log and predictions (2)

It can be proven that if $u \hookrightarrow N(0, \sigma^2)$, then $E[\exp(u_i)] = \exp(\frac{\sigma^2}{2})$. This can be estimated using:

$$\hat{\Psi} = \exp\left(\frac{\hat{\sigma}^2}{2}\right) \qquad \text{so that} \qquad \hat{y}_i = \exp(\hat{a} + \hat{b}x_i)\,\hat{\Psi}$$

To get back to levels when using a prediction in *logs*, we cannot simply take the exponential of the prediction: there is an additional retransformation parameter to take into account.
The preceding formula holds only if the u's are normal: if their distribution is unknown but homoskedastic (same variance), we can use Duan's non-parametric[1] *smearing* estimator:

$$\hat{\Psi} = \frac{1}{N}\sum_{i=1}^{N} \exp(\hat{u}_i)$$

[1] Because it makes no assumption on the type of distribution

Are all observations given the same weight?

We have:

$$\hat{b}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

So:

$$\hat{b}_1 = \frac{\sum(x_i - \bar{x})^2 \frac{(y_i - \bar{y})}{(x_i - \bar{x})}}{\sum(x_i - \bar{x})^2} = \sum p_i \frac{(y_i - \bar{y})}{(x_i - \bar{x})}$$

Calling $p_i = \frac{(x_i - \bar{x})^2}{\sum(x_i - \bar{x})^2}$.
$\frac{(y_i - \bar{y})}{(x_i - \bar{x})}$ is the slope of the line drawn from the point corresponding to individual i to the sample average, and p_i is an increasing function of $|x_i - \bar{x}|$.
Estimate $\hat{b}_1$ is thus highly influenced by points with extreme x values (see Anscombe's quartet: four datasets with the same regression line).

Outliers (1)

- Outlier: observation with a large residual compared to other observations
- They may indicate a poor goodness of fit
- How to detect them: using a residuals plot
- Goal: estimate how far the observed value is from the predicted value
- Or more rigorously: using the *studentized* residuals, which adjust for the *leverage* of that particular point

Outliers (2)

- Outliers are observations that have a large residual
- However, if the observation has a high leverage, it has pulled the predicted value towards itself
- In other words, the variances of the residuals differ, even though the variances of the true errors are all equal to each other
- Consequence: need for studentization
- The *studentized* residuals adjust for the leverage of that particular point (see what h is below)

$$rstud_i = \frac{\hat{u}_i}{\hat{\sigma}\sqrt{1-h_i}}$$

This is the *internally* studentized residual; the *externally* studentized one uses $\hat{\sigma}_{(-i)}$, estimated without observation i.

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Leverage (1)

- High leverage observation: has an explanatory variable that takes a value very different from the sample mean
- In a simple regression model, it could be (for example) at the far right and would influence the slope of the regression line
- How to detect them: the "hat statistic"

Leverage (2)

- We know that $\hat{y} = P_X y$
- P_X can be called the "hat" matrix (it puts a hat on y)
- Let's call h_i the ith element of the diagonal of the hat matrix
- h_i is the *leverage* of observation i: if it is large, then observation i has a high leverage
- $0 \leq h_i \leq 1$ (and $h_i \geq 1/N$ when the model has a constant)
- $\sum_i h_i = k$, so the average leverage is $\bar{h} = k/N$

Leverage (3)

- We know that $\hat{u} = (I - P_X)u$, so $V(\hat{u}) = \sigma^2(I - P_X)$
- So: $V(\hat{u}_i) = \sigma^2(1 - h_i)$
- The greater the leverage h_i, the smaller the variance of the residual for observation i
- Meaning: high leverage observations tend to bring the regression line close to them
- High leverage observations are not necessarily influential, but low leverage observations are rarely very influential
- For the simple linear model: $h_i = \frac{1}{N} + \frac{(x_i - \bar{x})^2}{\sum (x_i - \bar{x})^2}$

Influence measures (1)

- Influential observation: if removed from the regression, estimated parameters vary a lot
- How to detect them: the *Dfbeta* statistic
- For every variable k, its *Dfbeta* measures how much its coefficient would change if we removed one observation
- It is computed for every observation i and every coefficient k (an $N \times k$ table)
- The change in $\hat{b}_k$ if observation i is dropped is:

$$\text{Dfbeta}_{i,k} = \frac{\hat{b}_k - \hat{b}_{k,(-i)}}{\hat{\sigma}_{k,(-i)}}$$

with $\hat{b}_{k,(-i)}$ and $\hat{\sigma}_{k,(-i)}$ computed without observation i. It represents the number of standard errors by which $\hat{b}_k$ changes when observation i is removed.

Influence measures (2)

- *Dffit*: measures how much the prediction for observation i changes when the estimation is made *without* observation i
- $\text{Dffit}_i = \frac{\hat{y}_i - \hat{y}_{i,(-i)}}{\hat{\sigma}_{(-i)}\sqrt{h_i}}$
- *Covratio* statistic: measures the change in the determinant of the covariance matrix of the estimates when deleting the ith observation (indicates loss in precision)
- *Cook's distance*: measures the influence of observation i on the model as a whole, not on a single coefficient (it combines all the *Dfbeta*'s)
- $D_i = \frac{(\hat{b} - \hat{b}_{(-i)})'(X'X)(\hat{b} - \hat{b}_{(-i)})}{k\hat{\sigma}^2}$

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Cutoffs

- Leverage h_i of observation i: the average is $\bar{h} = k/N$. If $h_i > 2k/N$ (or $3k/N$), observation i has a high leverage and might be influential
- Change in predicted value *Dffit*: observation i is influential if $|\text{Dffit}_i| > 2$ (general cutoff) or $> 2\sqrt{k/N}$
- Change in a parameter *Dfbeta*: observation i is influential if $|\text{Dfbeta}_{i,k}| > 2$ (general cutoff) or $> 2/\sqrt{N}$ (adj. for large sample)
- Impact on the whole model (Cook's distance D): observation i is influential if $D_i > 1$ (some use $4/N$)
- *Covratio* statistic: observation i is influential if $|\text{covratio}_i - 1| > 3k/N$
- Python: `infl = res.get_influence()`; `infl.summary_frame()` gives leverage, Dfbeta, Dffit and Cook's distance for each observation, and `infl.cov_ratio` the covratio